

Representation of Data



Stem-and-leaf diagrams

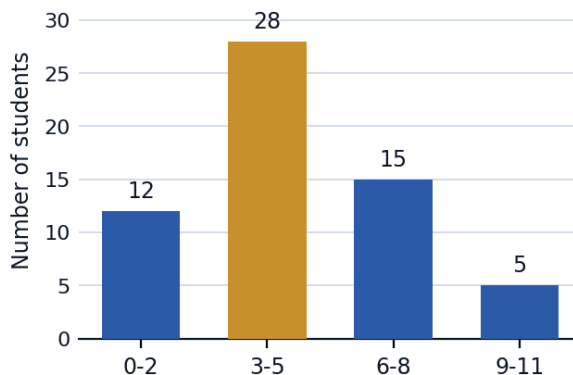
- 1 The data set 23, 25, 31, 34, 34, 38, 42, 45 is to be shown as a stem-and-leaf diagram with stems 2, 3, 4 . Write out the diagram, and use it to find the median.
- 2 The data set 17, 19, 22, 26, 26, 29, 33, 37, 41 is to be shown as a stem-and-leaf diagram with stems 1, 2, 3, 4. Write out the diagram, and use it to find the range.

Histograms and frequency density

- 3 A class interval of width 10 contains 25 items. Find the frequency density, which is plotted on the vertical axis of a histogram.
- 4 A class interval of width 4 contains 18 items. Find the frequency density, which is plotted on the vertical axis of a histogram.
- 5 A class interval of width 5 contains 30 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

Reading grouped frequency data from a bar chart

- 6 The bar chart shows the number of books read by students in a reading challenge, grouped by number of books. Find the total number of students, and the modal group.



- 7 Using the same reading-challenge data (0-2: 12, 3-5: 28, 6-8: 15, 9-11: 5 students), find what percent of students read 6 or more books.

Measures of central tendency for grouped data

- 8 A grouped data set has class midpoints and frequencies: 5 (freq 4), 15 (freq 8), 25 (freq 6), 35 (freq 2). Estimate the mean.
- 9 A grouped data set has class midpoints and frequencies: 10 (freq 3), 20 (freq 9), 30 (freq 5), 40 (freq 3). Estimate the mean.
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Box-and-whisker plots

- 10 A data set has minimum = 8, $Q_1 = 15$, median = 22, $Q_3 = 30$, and maximum = 40. State the interquartile range, and determine whether a value of 45 would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.
- 11 A data set has minimum = 5, $Q_1 = 12$, median = 18, $Q_3 = 27$, and maximum = 50. State the interquartile range, and determine whether the maximum value would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.
- 12 A data set has $Q_1 = 20$ and $Q_3 = 32$. Find the lower outlier boundary, defined as any value below $Q_1 - 1.5 \times IQR$.
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Skewness from summary statistics

- 13 A data set has mean = 42, median = 38, and mode = 35. Describe the skewness of the distribution.
- 14 A data set has mean = 28, median = 32, and mode = 35. Describe the skewness of the distribution.
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Comparing distributions using summary measures

- 15 Two classes take the same test. Class A has mean 68 and standard deviation 5. Class B has mean 71 and standard deviation 12. Compare the two classes' performance and consistency.
- 16 Two machines fill bottles with a target volume of 500 mL. Machine X has mean fill volume 500 mL and standard deviation 2 mL. Machine Y has mean fill volume 498 mL and standard deviation 8 mL. Compare the two machines' accuracy and consistency.
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A well-designed word problem: analyzing delivery times

- 17** This question has two parts. A delivery company records the delivery times (in minutes) for 50 packages: mean = 34 minutes, standard deviation = 6 minutes. Company policy states any delivery beyond 2 standard deviations above the mean is classified as late. (a) Find the delivery-time threshold, in minutes, above which a delivery is classified as late. (b) If a customer's package took 44 minutes to deliver, determine, using the standard deviation, whether this delivery should be classified as late, and state how many standard deviations from the mean this value lies.